

RESISTANCE, RESILIENCE, AND COMMUNITY RECOVERY IN INTERMITTENT RIVERS AND EPHEMERAL STREAMS

4.8

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- Intermittent rivers and ephemeral streams (IRES) are temporally and spatially dynamic, experiencing alternating wet and dry phases and supporting both aquatic and terrestrial habitats
- Persistence in these highly variable habitats requires aquatic taxa to be resistant or resilient to disturbances such as drying
- Resistance mechanisms include tolerance of extreme physicochemical conditions and having desiccation-resistant dormant stages; resilience mechanisms require the ability to disperse instream or overland
- Terrestrial taxa inhabiting dry riverbeds must be resistant or resilient to periodic flooding of their habitat
- Resistance and resilience mechanisms interact to facilitate community recovery
- Human disturbance alters recovery pathways and affects long-term persistence of aquatic and terrestrial species in IRES

4.8.1 INTRODUCTION: CHALLENGES OF PERSISTING IN IRES

Intermittent rivers and ephemeral streams (IRES) host complex mosaics of terrestrial and aquatic habitats that vary greatly through time and across landscapes (Chapters 2.2, 2.3, and 4.9). Some IRES flow for as little as several days or weeks each year, while others may flow nearly year-round (Fig. 4.8.1A). Furthermore, flow duration patterns can vary greatly across regions and years (Poff and Ward, 1989). Even within the same stream, adjacent reaches may have perennial flow, flow for several months, or flow only a few days annually (Jaeger and Olden, 2012). This dynamism in the location and duration of flowing, pool and dry habitats poses significant challenges to aquatic and terrestrial species. Despite

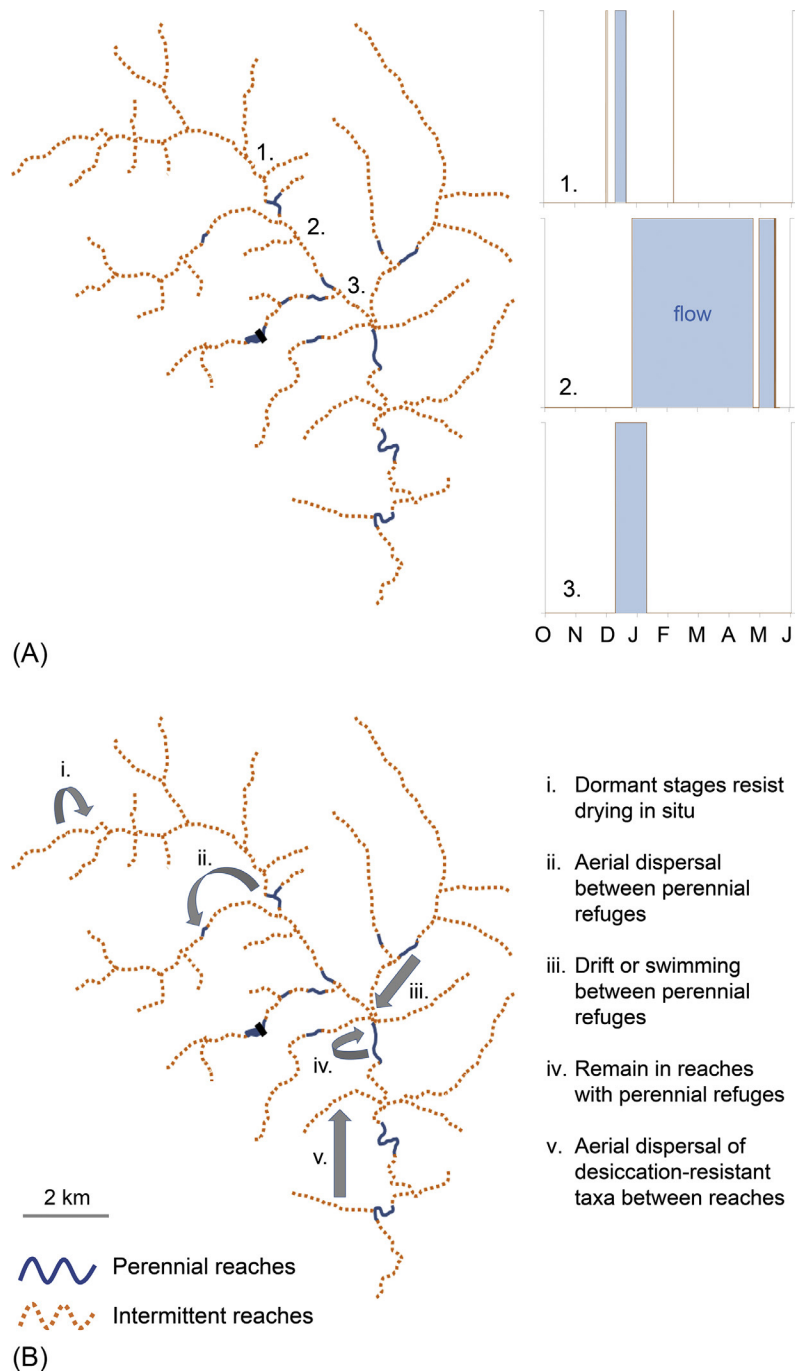


FIG. 4.8.1

Highly variable flow regimes across an IRES network in Mediterranean-climate California and potential dispersal pathways of aquatic organisms. In the top panel (A), three adjacent reaches in Chalone Creek (locations 1–3) exhibited varying periods of flow (shaded *blue*), lasting from a few weeks to several months, during the 2014–2015 wet season. In the bottom panel (B), five of the most common resistance and resilience mechanisms for aquatic species are illustrated. Importantly, these five strategies are not mutually exclusive as some species are capable of using multiple strategies. For example, the caddisfly *Lectrides varians* can use strategies i–v to persist in Australian IRES (Wickson et al., 2012).

these challenges, IRES can support diverse biological communities, including many species of algae, macrophytes, invertebrates, amphibians, reptiles, and fishes (Chapters 4.1–4.6). Although IRES experience numerous types of natural disturbances (e.g., fire, scouring floods), we focus here on the unique challenges that cycles of drying and rewetting present to resident taxa.

Freshwater and riparian taxa employ a wide range of strategies and traits to survive disturbances. Verberk et al. (2008) grouped these into four trait domains: dispersal, reproduction, synchronization, and development. This provides a framework that links the traits we observe to life histories and the phylogenies that constrain them. Traits used by taxa to survive cycles of wetting and drying in IRES exist in all four domains. For example, dispersal is used by many species to escape either wetting or drying (Section 4.8.3). Reproductive strategies used by animals in IRES vary greatly, from the rapid opportunistic production of large numbers of eggs by some desert fish (e.g., Minckley and Marsh, 2009) to long-lived crayfish that only reproduce in years with favorable conditions (e.g., Johnston et al., 2008). Synchronization refers to the timing of life history events such as dispersal, reproduction, and development, and may involve speeding up or slowing down development in response to environmental conditions (Verberk et al., 2008). Flexible development is often assumed to occur in IRES species, but little is known about the outcomes of this flexibility for recovery of stream communities.

Traits in each of these domains may comprise either resistance or resilience strategies for surviving disturbance. Here, we consider resistance and resilience to be distinct components of ecological responses to disturbance. Resistance is defined as the ability of an ecological unit (e.g., population, community) to withstand a disturbance *in situ* whereas resilience is defined as the capacity for an ecological unit to recover following disturbance (Lake, 2000; Nimmo et al., 2015). Aquatic species in many IRES are confronted with periodic loss of water and have various means of resisting, or being resilient to, stream drying (Box 4.8.1; Fig. 4.8.2). Terrestrial taxa must deal with the opposite situation—periodic inundation of their terrestrial habitat. When flow resumes, terrestrial taxa can be flushed downstream, experience high mortality or be forced to disperse laterally into riparian zones (Corti and Datry, 2012; Rosado et al., 2015). If terrestrial taxa cannot resist inundation, then they must crawl or fly overland to recolonize streambeds as flow ceases and aquatic habitat dries.

BOX 4.8.1 RESISTANCE, RESILIENCE, AND RECOVERY PROCESSES IN IRES

Species inhabiting IRES use a variety of *resistance* mechanisms to withstand disturbance *in situ* and *resilience* mechanisms to recover following disturbance (Lake, 2000; Nimmo et al., 2015). For example, many aquatic species persist in drying reaches by tolerating the harsh abiotic conditions in wetted microhabitats that remain after flow ceases (e.g., damp leaf packs, remnant pools; Figure 4.8.2; Chapter 4.3). Other aquatic species enter desiccation-resistant dormant stages during dry periods and reemerge as active individuals when flow returns (Williams, 2006).

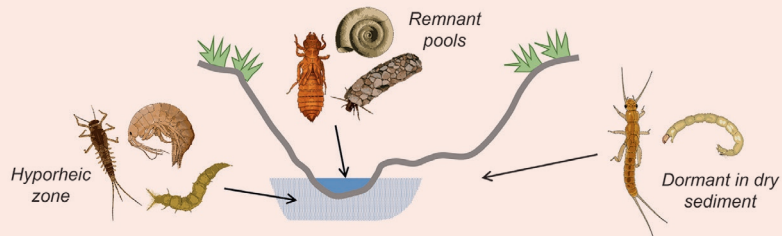
In contrast to resistance mechanisms that allow individuals to persist through a disturbance, resilience mechanisms facilitate repopulation of disturbed reaches from elsewhere in the catchment or neighboring catchments. For example, most aquatic insect species have adult stages capable of flight, so adults from perennial streams can fly to, and lay eggs in, neighboring IRES when flow resumes (Figs. 4.8.1B and 4.8.2). Other taxa, such as fish, may swim rapidly through IRES networks during brief periods of flow connectivity (Chapter 4.5), allowing individuals to disperse from distant perennial refuges and recolonize intermittent reaches (Marshall et al., 2016; Datry et al., 2016a; Chapter 4.9).

Both resistance and resilience contribute to *recovery*, considered here to be a return to predisturbance levels of any ecological unit of interest to researchers or managers (e.g., taxonomic diversity, abundance, community composition; Nimmo et al., 2015).

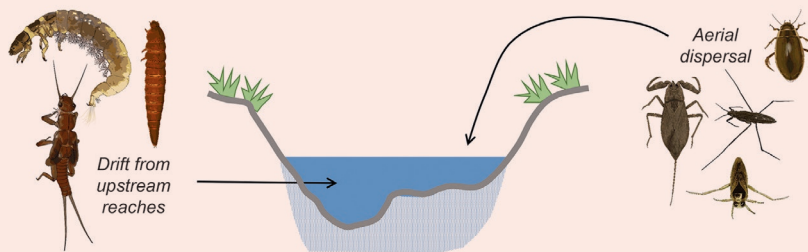
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BOX 4.8.1 RESISTANCE, RESILIENCE, AND RECOVERY PROCESSES IN IRES (Cont'd)**RESISTANCE**

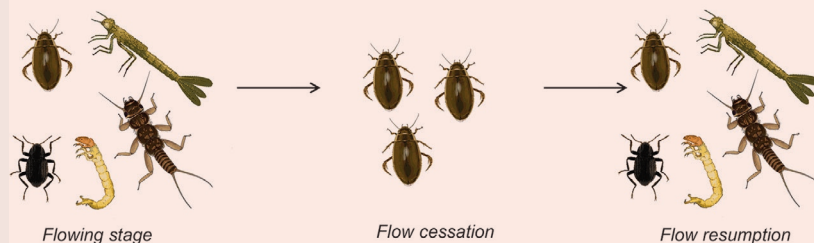
ability to survive flow cessation and persist in a local reach

**RESILIENCE**

ability to recolonize a local reach from elsewhere after flow resumes

**RECOVERY**

population or community returns to its *predisturbance* state

**FIG. 4.8.2**

Definitions and illustrations of resistance, resilience and recovery as used in this chapter. Although aquatic invertebrates are used here as examples, similar patterns are seen with other aquatic taxa (e.g., algae, fish). Terrestrial communities inhabiting dry riverbeds also exhibit resistance and resilience to inundation, and may demonstrate parallel patterns of community recovery following surface water loss.

Invertebrate drawings: Courtesy P. Fortuño Estrada.

In this chapter, we first explore resistance and resilience mechanisms that aquatic taxa use to persist locally in IRES during flow cessation, including complete stream drying. We then describe how the resistance and resilience mechanisms of individual species scale up to facilitate community recovery to predisturbance levels of diversity, abundance, and composition. Although much remains to be learned about terrestrial invertebrates in dry riverbeds ([Chapter 4.4](#)), we summarize what is known about how terrestrial taxa resist or are resilient to inundation in IRES. We also highlight how resistance and resilience mechanisms shape the landscape genetics of IRES taxa. Finally, we discuss how human disturbances can interfere with resistance and resilience pathways and conclude by suggesting several future research directions.

4.8.2 RESISTANCE MECHANISMS OF AQUATIC TAXA

When flow ceases in IRES, aquatic taxa must disperse away from drying reaches ([Section 4.8.3](#)) or persist locally. Organisms that can survive dry seasons or years within a given IRES reach are termed “resistant” to the effects of flow cessation. To survive flow cessation, however, organisms must tolerate the often harsh abiotic conditions in remaining wetted habitats or be capable of entering a desiccation-resistant stage. Examples of standing water or damp habitats that can persist after surface flow ceases include remnant pools, crayfish burrows, leaf packs, algal mats, and the hyporheic zone (i.e., saturated sediments below the stream channel) ([Chapter 4.3](#)). However, these habitats lack swiftly flowing water and each type presents challenges for aquatic taxa. Even the survival of desiccation-resistant stages is reduced during especially harsh or long dry periods ([Storey and Quinn, 2013](#); [Stubbington and Datry, 2013](#)). Three of the most common strategies that aquatic fauna use to resist drying in IRES are surviving in remnant pools, taking refuge locally in the hyporheic zone, and entering desiccation-resistant dormant stages.

REMNANT POOLS

As flow ceases, aquatic organisms in IRES are often concentrated into remnant pools ([Boulton and Lake, 1992](#); [Bogan and Lytle, 2007](#); [Chapter 4.3](#)). This concentration effect can result in high densities and diversity of species in relatively small pools ([Fig. 4.8.3](#)). In fact, taxa richness in remnant pools during dry periods often equals that of riffles during flowing periods in desert and Mediterranean-climate streams ([Bonada et al., 2006](#); [Bogan and Lytle, 2007](#); [Driver and Hoeinghaus, 2016](#)). Remnant pools also can serve important roles beyond local reaches, acting as biodiversity reservoirs for entire drainage networks ([Sheldon et al., 2010](#); [Chester and Robson, 2011](#); [Robson et al., 2013](#)). However, remnant pools can also be harsh environments for aquatic organisms. For example, these pools frequently exhibit low dissolved oxygen concentrations and high water temperatures ([Boulton et al., 1992](#)). Significant daily fluctuations in these and other water quality variables ([Drummond et al., 2015](#); [Datry, 2017](#)) can greatly reduce the survival of aquatic taxa ([Driver and Hoeinghaus, 2016](#)). Furthermore, some IRES lack remnant pools, and severe drought may cause formerly perennial pools to dry ([Bogan and Lytle, 2011](#)).

Taxa vary greatly in their use of remnant pools to resist flow cessation. Individuals of some algae and diatom species remain “active” in remnant pools and help repopulate intermittent reaches when flow returns ([Robson et al., 2008](#)). Aquatic invertebrates frequently use remnant pools to survive dry

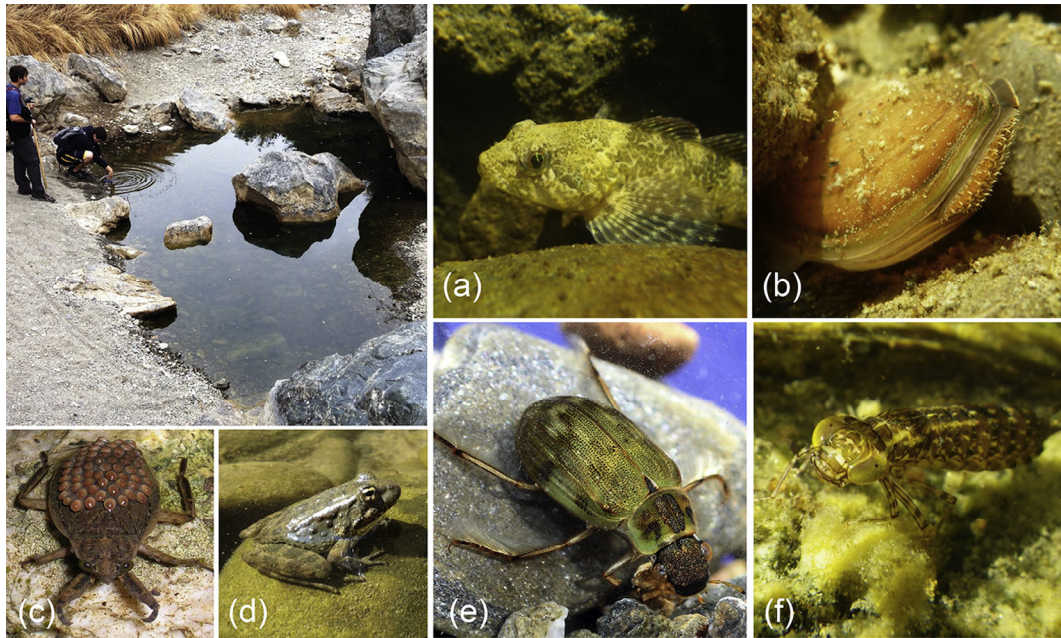


FIG. 4.8.3

Refuge pool at Coyote Creek, California, USA and six of the >80 aquatic taxa that inhabited the refuge pool in October 2014: (a) riffle sculpin (*Cottus gulosus*), (b) California floater mussel (*Anodonta californiensis*), (c) giant water bug (*Abedus* sp.), (d) foothill yellow-legged frog (*Rana boylii*), (e) water scavenger beetle (*Berosus* sp.), and (f) damselfly nymph (*Aeshna* sp.).

Photos: Courtesy M. Bogan (a–d, and f) and M. Cover (e).

periods (Boulton and Lake, 2008; Chester and Robson, 2011; Chapter 4.3), especially air-breathing taxa such as adult beetles and true bugs (Bonada et al., 2006; Bogan and Lytle, 2007; Fig. 4.8.3). Pools are especially important for taxa that cannot disperse aurally, such as fish, crustaceans, and amphibians (e.g., Labbe and Fausch, 2000). Many native fishes in IRES are adapted to tolerate the harsh abiotic conditions in remnant pools (Chapter 4.5), including high temperatures, low pH and dissolved oxygen concentrations, and high dissolved organic carbon concentrations (McMaster and Bond, 2008; Driver and Hoeinghaus, 2016). For example, the Australian desert goby can tolerate water temperatures of up to 41°C and salinity levels twice that of seawater (Glover, 1982; Thompson and Withers, 2002). Minnows in Sonoran Desert streams (USA) burrow into wet algal mats as pools dry during daytime heat and then resume activity at night when water levels increase (Minckley and Barber, 1971). Even fish that are traditionally considered sensitive, such as salmonids, can thrive in remnant pools (Wigington et al., 2006; Hwan and Carlson, 2016). While many amphibian species are adapted to exploit seasonal and ephemeral waters (Chapter 4.6), species that cannot estivate in terrestrial habitats require remnant pools to persist in IRES (e.g., yellow-legged frogs in the United States: Fig. 4.8.3; Stebbins, 2003).

HYPORHEIC ZONE

When remnant pools are lost and surface channels dry completely in gravel-bed streams, many taxa passively or actively find refuge in the hyporheic zone (Del Rosario and Resh, 2000; DiStefano et al., 2009; Kawanishi et al., 2013). During dry periods, this zone may offer more stable abiotic conditions than at the surface, and thus host more diverse biotic communities (Stubbington, 2012). In some systems, such as the Eygues River in France, the most common benthic invertebrate taxa are also found in hyporheic zones during brief periods of surface drying, and form the primary source of recolonists when flow returns (Vander Vorste et al., 2016a). Hyporheic refuge use is not limited to invertebrates; some fish persist locally by entering the hyporheic zone when surface water dries (Kawanishi et al., 2013). However, interstitial architecture affects hyporheic refuge use, with fine sediments and small pores excluding large individuals (Gayraud and Philippe, 2001; Descloux et al., 2013; Vadher et al., 2015). Additionally, the physiological condition of individuals can decline with time spent in the hyporheic zone (e.g., fish: Kawanishi et al., 2013). Furthermore, continued drought conditions eventually cause hyporheic water levels to decline. While some species (e.g., crayfish: Flinders and Magoulick, 2003) can actively track falling water tables, doing so may come at a cost. For example, amphipods in IRES can follow declining water tables but their survival may be lower at greater depths (Vander Vorste et al., 2016b).

DESICCATION-RESISTANT STAGES

Having a desiccation-resistant life stage is the only viable resistance strategy when all wetted surface and hyporheic refuges are absent. Primary producers, including chlorophytes and cyanobacteria, commonly use dormant stages to resist drying (Chapter 4.2), becoming active again within hours or days of flow resumption (Robson et al., 2008; Timoner et al., 2014). Aquatic invertebrate taxa may have one or more desiccation-resistant stages within their life history, including resistant eggs, larvae, pupae, or adults (Strachan et al., 2015; Chapters 4.3 and 4.10). “Seedbanks” of active and dormant invertebrates have been widely documented from dry streambeds (Larned et al., 2007; Stubbington and Datry, 2013; Strachan et al., 2015). For example, rehydration experiments conducted in New Zealand and France found that >50% of flowing-phase taxa were found in rehydrated sediments collected from dry gravel-bed streams, suggesting that invertebrate seedbanks can be diverse (Datry et al., 2012; Storey and Quinn, 2013). Larvae of the dobsonfly *Neohermes filicornis* can survive multiple drying and rewetting cycles and may take up to 5 years to reach maturity in IRES with short flow durations (Cover et al., 2015). Additionally, some crayfish and isopod species construct small estivation chambers beneath stones (Fig. 4.8.4). A few vertebrate taxa are also capable of resisting complete stream-drying events. For example, African lungfish can burrow into the mud as streams dry, secrete a cocoon, and lie dormant for up to four years until water returns (Greenwood, 1987). However, reduced streambed moisture can lower the survival rates of active and dormant taxa (Wickson et al., 2012; Stubbington and Datry, 2013; Storey and Quinn, 2013). Furthermore, these desiccation-resistance strategies may not be useful in all IRES. For example, desiccation-resistant invertebrates are rarely found in dry headwater and bedrock streams dominated by coarse substrata or mobile sands (Fritz and Dodds, 2004; Chester and Robson, 2011).

**FIG. 4.8.4**

Examples of aquatic invertebrates that estivate under stones in dry or damp streambeds to resist drying in some Australian IRES: (a) the crayfish *Geocharax* sp. nov. 1 with stone removed to show estivation chamber, (b) isopods (*Synamphisopus doegi*, as seen below the white arrows) which estivate in damp sand under stones, and (c) a dragonfly nymph (Telephlebiidae) that was found under a large stone.

Photos: Courtesy B. Robson.

4.8.3 RESILIENCE MECHANISMS OF AQUATIC TAXA

If individuals are unable to persist locally when flow ceases, then resilience mechanisms become essential for maintaining populations of aquatic taxa in IRES (Fig. 4.8.2). Resilience via dispersal is frequently cited as the most important means of achieving recovery in IRES (Acuña et al., 2005; Sheldon et al., 2010; Chester and Robson, 2011). The ability to disperse among wetted reaches, including via overland flight, drift dispersal, and swimming, is widely documented among IRES taxa (e.g., Williams and Hynes, 1976; Robson et al., 2008; Lake, 2011). These forms of spatial dispersal allow organisms to recolonize a previously dry reach from elsewhere when flow returns. In fact, many aquatic taxa in IRES can only maintain local populations via spatial dispersal from elsewhere in the catchment or adjacent catchments (e.g., Stanley et al., 1994; Chester et al., 2014). In the following sections, we discuss several commonly used dispersal pathways that confer resilience to IRES populations and explore some of the cues that trigger dispersal away from drying habitats.

DISPERSAL VIA DRIFT

When perennial reaches are located upstream of temporary reaches in IRES, organisms can passively disperse via drift to recolonize downstream reaches. For example, after flow resumes, diatoms rapidly enter the drift or raft on floating detritus, colonizing downstream reaches and influencing assemblage composition (Robson et al., 2008). Drift is especially well studied with regard to aquatic invertebrates (e.g., Larimore et al., 1959; Williams and Hynes, 1976) and certain insect taxa are known to be “drift prone” (e.g., baetid mayflies). Several studies, including those from IRES in tropical Australia and montane temperate North America, have identified drift as being the primary source of invertebrate colonization when flow returns (McArthur and Barnes, 1985; Paltridge et al., 1997). However, in the short term (days), drift in gravel-bed rivers may not contribute as many colonists as emergence from the hyporheic zone (Vander Vorste et al., 2016a). Larval drift can also be an important dispersal mechanism for amphibians, like the fire salamander of Israeli ephemeral streams (Segev and Blaustein, 2014). Drift dispersal of eggs and larvae helps to maintain fish populations in perennial rivers (e.g., Copp et al., 2002; Minckley and Marsh, 2009) and is also likely to promote population resilience in IRES.

DISPERSAL VIA CRAWLING AND SWIMMING

Invertebrates that are not prone to entering the drift, such as heptageniid mayflies, may recolonize IRES by crawling from upstream or downstream perennial reaches when flow resumes (McArthur and Barnes, 1985). In Canadian IRES, midge larvae, nematodes, oligochaetes, ostracods, cyclopoids, and amphipods have been observed crawling and swimming upstream into intermittent reaches from downstream perennial reaches (Williams, 1977). In some desert IRES, flightless giant water bugs (Belostomatidae) crawl and swim through wetted reaches when flow resumes and can even crawl short distances through dry stream channels or terrestrial habitats (Lytle, 1999; Boersma and Lytle, 2014). Although fish are generally limited to instream dispersal pathways, many species disperse quickly through IRES networks when flow returns (Chapter 4.5). In Australia, for example, several fish species swim 70 km or more between isolated waterholes when ephemeral rivers experience flooding (Kerezy et al., 2013; Marshall et al., 2016). Additionally, fish in IRES may disperse further or more frequently than those from perennial habitats, thus enhancing population resilience following flow resumption, as

seen in some Mediterranean streams (Pires et al., 2014). In temperate IRES of the northwestern United States, fish move into intermittent reaches to spawn, and these reaches also act as nurseries for juveniles (Colvin et al., 2009). However, fish species richness decreases with distance from perennial waters, suggesting limitations on dispersal for some species (Colvin et al., 2009).

AERIAL DISPERSAL

Aerial adult life stages of aquatic insects facilitate dispersal both within and between catchments (Figs. 4.8.1B and 4.8.5), providing an important source of colonists in IRES. Adult flight has been cited by some as being the principal recolonization mechanism for many species of aquatic insects (Gray and Fisher, 1981; Abell, 1984). This may especially be the case in dryland regions, where hydrological connections to perennial waters are rare (Davis et al., 2013; Cañedo-Argüelles et al., 2015; Razeng et al., 2016). In fact, genetic studies have demonstrated that entire populations at the reach scale can be the result of just one or two matings of adult aquatic insects (Bunn and Hughes, 1997), highlighting the importance of aerial dispersal in maintaining local populations.

While many aquatic insects (e.g., true flies, mayflies) are relatively weak fliers, others are quite capable of dispersing many kilometers through the air, including beetles, dragonflies, and true bugs (Fig. 4.8.5). Although aerial dispersal abilities vary greatly from species to species, colonization rates of isolated habitats generally decline with distance from source populations as flight capacities are exceeded (Bogan and Boersma, 2012; Chester et al., 2015). Weak dispersers (e.g., flightless giant water bugs, water penny beetles) may be lost from communities when dry period durations and dry reach extents are increased by drought (Bogan and Lytle, 2011; Chester et al., 2015). However, long reaches

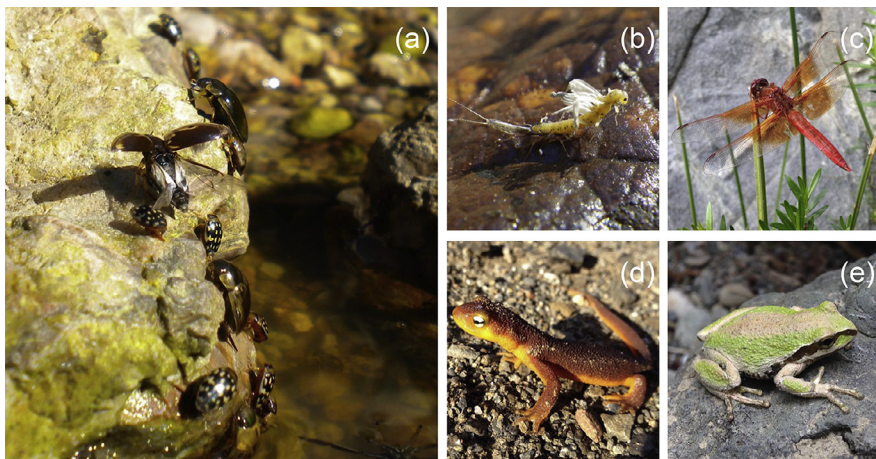


FIG. 4.8.5

Overland dispersal of aquatic taxa: (a) water scavenger and predaceous diving beetles (Hydrophilidae and Dytiscidae) escaping a drying stream pool, (b) winter stonefly (*Capnia californica*) emerging into aerial adult stage, (c) flame skimmer dragonfly (*Libellula saturata*) adult, (d) California newt (*Taricha torosa*) dispersing through upland terrestrial habitat, and (e) Sierran treefrog (*Pseudacris sierra*) dispersing along dry stream channel.

Photos: Courtesy M. Bogan.

of dry streambeds are not necessarily impediments to aerial dispersal and may even serve as dispersal highways for flying aquatic insect adults (Bogan and Boersma, 2012; Morán-Ordóñez et al., 2015). Many flying insects navigate using visual cues, such as polarized light from water and smooth stones, and ultraviolet light from plants (Collett, 1996; Kriska et al., 2009), which may be restricted to streambeds in arid regions. Dispersal between streams may also be facilitated by concave topographies, such as depressions between sand dunes, which may contain similar visual cues due to water accumulation during rain events (Razeng et al., 2016). During rain events, small, ephemeral pools can form in depressions outside the river network and offer a “stepping-stone” route between streams (Bunn et al., 2006).

ENVIRONMENTAL CUES THAT TRIGGER DISPERSAL

Aquatic taxa use various environmental cues to know when to escape drying reaches or swim through reconnected IRES during flow pulses. Some fish and invertebrates crawl or swim upstream as flow recedes in intermittent reaches (positive rheotaxis: Davey and Kelly, 2007; Lytle et al., 2008). Declining water levels in pools can prompt aquatic insects to crawl out of the water and disperse short distances overland in search of other remnant pools (Fig. 4.8.5; Boersma and Lytle, 2014). Large mixed flocks of aquatic beetle and true bug species have been observed dispersing aerially in dryland regions, often as a result of mass emigration from drying habitats (Kingsley, 1985; Stevens et al., 2007). Indirect drying cues, such as increasing water temperature, may also trigger aerial dispersal of aquatic insects (Velasco and Millan, 1998). Heavy rainfall and associated high-velocity flow conditions can trigger fish to disperse through IRES networks, allowing them to colonize rewetted reaches or move among remnant pools during brief flow events (Labbe and Fausch, 2000; Pires et al., 2014).

4.8.4 AQUATIC COMMUNITY RECOVERY PATTERNS FOLLOWING FLOW RESUMPTION

Aquatic communities in IRES with natural flow regimes tend to recover quickly following flow resumption, but exact timing of recovery varies by taxonomic group and region. Even within a region, recolonization potential is likely to vary with physical connectivity, flow regime, occurrence of perennial refuges, and other environmental factors (Baguette et al., 2013). Despite this variability, biofilms are often the first component of IRES communities to fully recover following flow resumption (Larned et al., 2007; Chapter 4.1). Although full recovery of algal biofilms usually occurs within 12 weeks of flow resumption, the length of the preceding dry period and flow conditions during the previous flow period can alter recovery timing and assemblage trajectories (Robson et al., 2008; Chester and Robson, 2014; Timoner et al., 2014). In some IRES, longer dry periods are associated with higher densities of cyanobacteria on stream stones (Robson et al., 2008). As cyanobacteria are seldom palatable to stream grazers, increases in drying duration may reduce food supplies to algal grazers in IRES (Robson et al., 2008), thereby slowing food web recovery following drying.

Aquatic invertebrate assemblages exhibit great variability in recovery times (Box 4.8.2), but recovery generally occurs within months. Rapid recovery is enabled by both resistance and resilience processes (e.g., Chester and Robson, 2011). Densities can recover in as little as one month, but species richness may take several months longer, especially in isolated sites (Boulton and Lake, 1992; Stanley et al., 1994). In the western United States, for example, IRES invertebrate assemblages are composed

BOX 4.8.2 EXAMPLES OF RECOVERY TRAJECTORIES IN IRES

Aquatic invertebrate communities exhibit a wide range of recovery patterns following flow resumption that are shaped by drying intensity, local habitat conditions, and the spatial configuration of wetted refuges (Chapters 4.3 and 4.9), among other factors. Despite this variability, species richness generally increases through time, with some species returning quickly and others colonizing later. Here, we illustrate how species richness recovers and community composition develops in two systems where these processes have been well documented: temperate alluvial rivers and desert streams.

In alluvial rivers with relatively short dry periods ranging from a few days to a couple of months, numerous species (e.g., mayflies, caddisflies, true flies, amphipods) seek refuge in the hyporheic zone during drying but rapidly return to the surface when flow resumes (Fig. 4.8.6A; Vander Vorste et al., 2016b). Thus, species richness increases quickly in these systems when flow resumes. Additional taxa (e.g., other mayflies and caddisflies, riffle beetles) soon arrive via drift from upstream reaches. Later, aerial colonization occurs via oviposition (e.g., dragonflies and caddisflies), and nonflying taxa (e.g., snails) arrive via crawling or other forms of instream dispersal (Fig. 4.8.6A; Williams, 2006).

In contrast, intermittent desert streams with long dry periods (>6 months/year) and no perennial headwaters tend not to support diverse hyporheic refuges (Bogan et al., 2015). In these systems, species richness recovers slowly. Desiccation-resistant stonefly, blackfly, and midge taxa are the first pioneers following flow resumption (Fig. 4.8.6B; Bogan et al., 2013). After a couple of months, several other taxa may colonize via adult dispersal and oviposition (e.g., mayflies, diving beetles). With longer flow duration, aerial colonization continues by active and passive or phoretic dispersers (e.g., caddisflies and snails, respectively) and a few species may colonize via instream movement from downstream perennial reaches (e.g., adult riffle beetles; Fig. 4.8.6B). Recent work quantifying recovery patterns in little studied IRES (e.g., Neotropical IRES: Datry et al., 2016b) will greatly aid efforts to synthesize recovery patterns across diverse biomes.

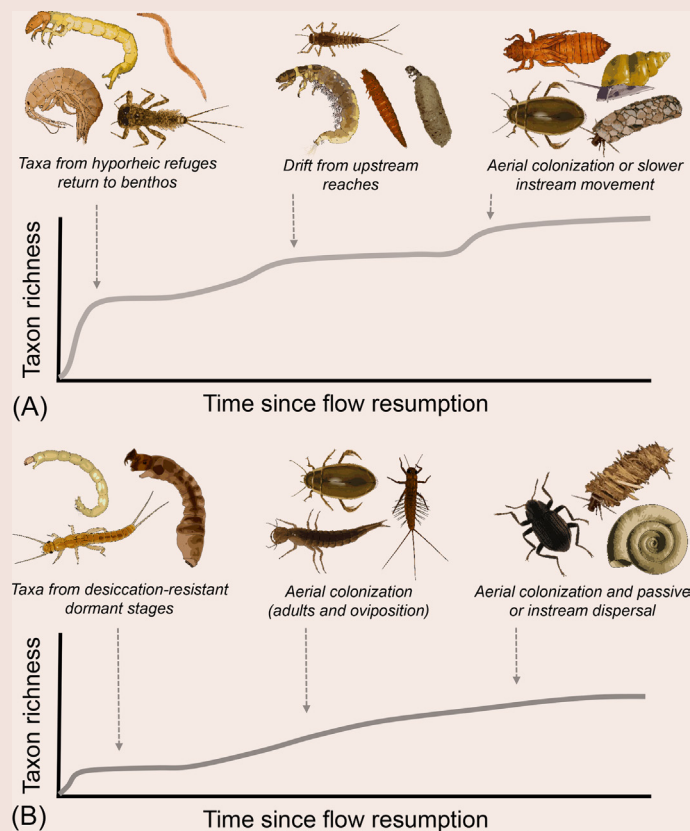


FIG. 4.8.6

Examples of aquatic invertebrate community recovery trajectories in (A) temperate gravel-bed and (B) desert IRES.

Invertebrate drawings: Courtesy P. Fortuño Estrada.

primarily of desiccation-resistant stonefly, blackfly, and midge taxa in the first three months of flow (Bogan et al., 2013). However, longer hydroperiods (e.g., 5–6 months) allow more time for resilient taxa, such as mayflies and dragonflies, to colonize intermittent reaches and promote recovery to predrying assemblage composition (Fig. 4.8.6; Bogan et al., 2015). Recovery in highly isolated (e.g., >10 km from perennial refuges) or hydrologically disjunct sites may take multiple years, if it happens at all (Bogan and Lytle, 2011). Because many adult aquatic insects preferentially fly along stream corridors and over short distances (a few kilometers: Briers et al., 2004; Bogan and Boersma, 2012; Chester et al., 2015), long-distance aerial dispersal events that facilitate colonization of isolated IRES may occur infrequently. Some studies report distinct invertebrate assemblages in the first few months after flow resumption, and these compositional differences may be larger in dry than in wet years (Abell, 1984; Acuña et al., 2005). However, full recovery to predrying community composition tends to happen within several months after flow resumption. Studies from both temperate and tropical regions have documented significant recovery in as little as 3 weeks (Datry, 2012; Datry et al., 2016b).

Fish assemblages often take longer to recover from dry periods in IRES because fish species generally rely on hydrological connectivity to disperse (except amphibious species: Clayton, 1993). Dispersal capacity varies greatly among fish species, with some dispersing rarely (Hughes et al., 2012) and other dispersing frequently when the opportunity arises (Bostock et al., 2006). Frequent drying events slow recolonization by most fishes and can result in IRES hosting fish assemblages distinct from those in perennial reaches (Davey and Kelly, 2007). Fish assemblage recovery also depends on the timing of drying; when drying and spawning periods coincide or drying occurs immediately after spawning, recovery can be severely impaired (Detenbeck et al., 1992). Recolonization of previously dry reaches is often a highly ordered process, resulting in temporal nestedness of assemblages in the months following flow resumption (Taylor and Warren, 2001; Whitney et al., 2016). Species that were the most abundant or widely distributed before stream drying, or that have traits such as small body sizes and flexible spawning habits, tend to be first to recolonize rewetted reaches (Detenbeck et al., 1992; Whitney et al., 2016). In contrast, fish species that are larger or take longer to reach maturity may take longer to recover.

Although resilience mechanisms are often reported as being the primary means of community recovery in IRES (e.g., Paltridge et al., 1997; Labbe and Fausch, 2000; Chester and Robson, 2011), both resistance and resilience play a role. Furthermore, the relative importance of their roles varies with local environmental conditions and across biomes. For example, studies of large alluvial rivers have found that hyporheic invertebrates, and resistance in general, contribute greatly to community recovery (e.g., Larned et al., 2007; Datry et al., 2012). In contrast, many bedrock-dominated headwater IRES lack hyporheic communities and rely more on resilience traits for faunal recovery (e.g., Bogan et al., 2015; Chester et al., 2015). Biome and biogeography also play a role, with dryland IRES tending to support more taxa with desiccation-resistance traits than wetter biomes, and thus resistance makes a larger contribution to recovery in those IRES (e.g., Leigh et al., 2016).

4.8.5 DYNAMICS OF TERRESTRIAL COMMUNITIES INHABITING DRY STREAMBEDS

When streambeds dry in IRES (Fig. 4.8.7), new habitat opens up for terrestrial and riparian species and terrestrial successions are initiated (Steward et al., 2011; Corti et al., 2013). When drying commences, riparian invertebrates disperse *en masse* from adjacent riparian zones to benefit from the large quantities

**FIG. 4.8.7**

IRES transitioning from aquatic to terrestrial habitat. Drying sequences are illustrated for a bedrock and a gravel-bed stream, from high-flow spates (*top*) to declining flows (*middle*), and flow cessation (*bottom*). In the bedrock stream (a–c: Cultivation Creek, Victoria, Australia), remnant pools persist through the dry season. However, in the gravel-bed stream (d–f: Chalone Creek, California, USA), no surface water remains during the dry season.

Photos: Courtesy E. Chester (a–c) and M. Bogan (d–f).

of food left over from the aquatic phase. An initial “clean-up crew” (Abell, 1984) of amphibious and terrestrial organisms consumes the stranded aquatic biota (Fig. 4.8.8), including dead and dying fish, aquatic invertebrates, and macrophytes (Williams, 2006). Some may stay in the dry riverbeds for long periods of time, such as lycosid spiders or carabid beetles (Paetzold et al., 2006; Greenwood and McIntosh, 2010; Datry et al., 2014). Others that require shade or vegetation, such as ants or springtails,

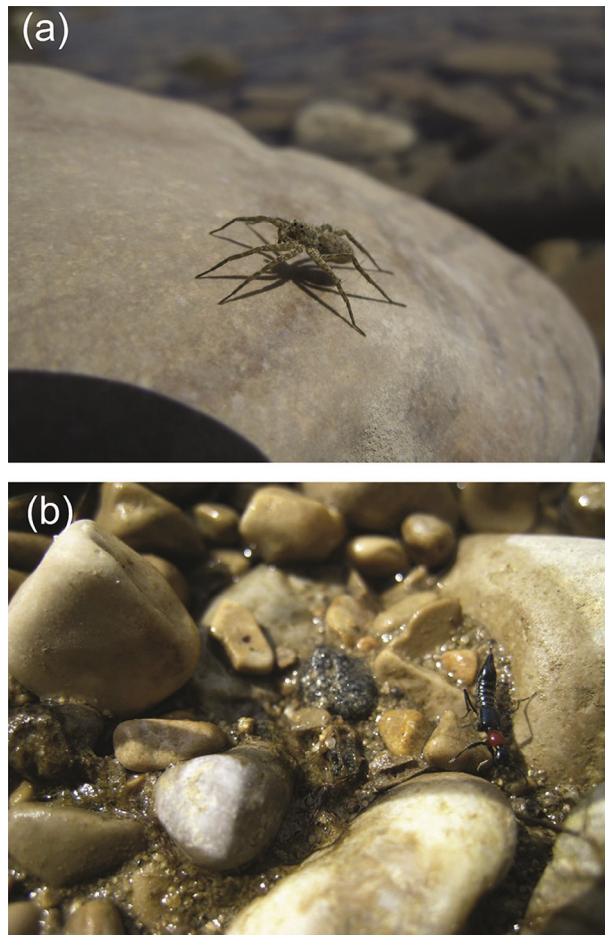


FIG. 4.8.8

Two common arthropods living along intermittent river shorelines and in dry riverbeds after flow ceases are (a) wolf spiders (Lycosidae) and (b) rove beetles (Staphylinidae). The wolf spider *Pardosa wagleri* is from the Asse River, France, and the rove beetle *Paederidus ruficollis* is from a drying reach of the Albarine River, France.

Photos: Courtesy R. Corti.

may use the dry riverbeds only temporarily. Terrestrial communities that develop in dry streambeds can be highly diverse and can differ from adjacent riparian communities (Steward et al., 2011) or just be a subset (Corti and Datry, 2016). As the duration of the dry period increases, streambeds become increasingly terrestrial and fewer aquatic species are able to survive in desiccation-resistant forms (Larned et al., 2007; Datry et al., 2012; Herschkovitz and Gasith, 2013). Despite the recent surge in research regarding the terrestrial phases of IRES, these systems remain poorly understood (Datry et al., 2011, Steward et al., 2012; Chapter 4.4).

During dry phases, IRES can act as dispersal highways, and hence resilience pathways, for terrestrial organisms (Coetzee, 1969; Chapter 4.6). Dry riverbeds contain fewer obstructions (e.g., trees, roads) than adjacent terrestrial habitats and can facilitate an organism's movements on the ground or through the air (Steward et al., 2012; Sánchez-Montoya et al., 2015). In dryland IRES, riparian areas can be more forested than adjacent habitats and this may favor dispersal by terrestrial organisms that are sensitive to desiccation, such as adult aquatic insects (Bogan and Boersma, 2012). In contrast, conditions in dry riverbeds in other regions can be very harsh with strong winds, intense solar radiation, and surface temperatures exceeding 60°C (Steward et al., 2011; Corti et al., 2013). This may limit the daytime movements of some terrestrial organisms along IRES.

When flow resumes, large numbers of terrestrial organisms can be entrained downstream in advancing wetting fronts (Corti and Datry, 2012; Rosado et al., 2015; Fig. 4.8.9), while other terrestrial species escape into the riparian zone (Boumezzough and Musso, 1983; Lambeets et al., 2008). Entrained organisms provide a food source for scavengers and predators from aquatic food webs, but some terrestrial organisms may “surf” these advancing fronts to colonize downstream riparian zones (e.g., some ants, beetles, and spiders: Corti and Datry, 2012; Rosado et al., 2015; Chapter 4.9). This passive dispersal method may be particularly effective for weak dispersers, such as springtails and spiders, and therefore be crucial for maintaining terrestrial invertebrate diversity along IRES corridors (Steward et al., 2012; Rosado et al., 2015).

4.8.6 GENETIC IMPLICATIONS OF DIFFERENT MODES OF RESISTANCE AND RESILIENCE

If resilience is the main recovery pathway in IRES, then most species should be strong dispersers, maintaining large populations with widespread genetic mixing (i.e., panmixia). However, the wide variety of resistance and resilience traits and the combinations of these traits actually observed in IRES show that patterns of population structure among species are far more complex (Murphy et al., 2010; Chester et al., 2015; Phillipsen et al., 2015). Studies of species with different lineages occupying the same landscapes have shown that both phylogenetic and life history differences among IRES species drive population structure (Murphy et al., 2010). Habitat plays a secondary role for species with specific habitat requirements (Chester et al., 2015; Phillipsen et al., 2015). Clearly, not all species are good dispersers, even in streams experiencing frequent disturbances that cause high rates of local extinction. This is an important conceptual advance that has come from research in IRES and emphasizes that present-day habitat conditions (i.e., cycles of drying and rewetting) have not restricted the variety of life histories or population structures in IRES.

Many IRES show dramatic expansions and contractions of habitat, during which some species disperse along river channels (and across floodplains, if present) and others do not (Sheldon et al., 2010). Resilient species that disperse and reproduce rapidly may form large populations with low genetic structure across landscapes (e.g., spangled perch *Leiopotherapon unicolor*, Bostock et al., 2006). In contrast, resistant species that do not disperse widely may show strong geographic structure in genetic patterns, comprising distinct populations that rarely interbreed (e.g., crayfish *Geocharax* sp. nov. 1, Chester et al., 2015; Fig. 4.8.4; Chapter 4.10). We might expect that these two patterns of population structure would dominate IRES species. However, some IRES species possess both resistance and

**FIG. 4.8.9**

Examples of dry streambed and subsequent flow resumption in four IRES: Bear Gulch, California, USA (a, b), Tea Tree Creek, Victoria, Australia (c, d), John West Fork, California, USA (e, f), and Gila River, New Mexico, USA (g, h).

Photos: Courtesy E. Chester (a–d), M. Bogan (e, f), and J. Whitney (g, h).

resilience traits. For example, the caddisfly *Lectrides varians* has adults capable of dispersal over distances ≥ 100 km (resilience), and larvae capable of estivation (resistance), but shows panmixia (Chester et al., 2015). Hence both resistance and resilience traits contribute to species strategies for surviving the “bust” and exploiting the “boom” (Mossop et al., 2015).

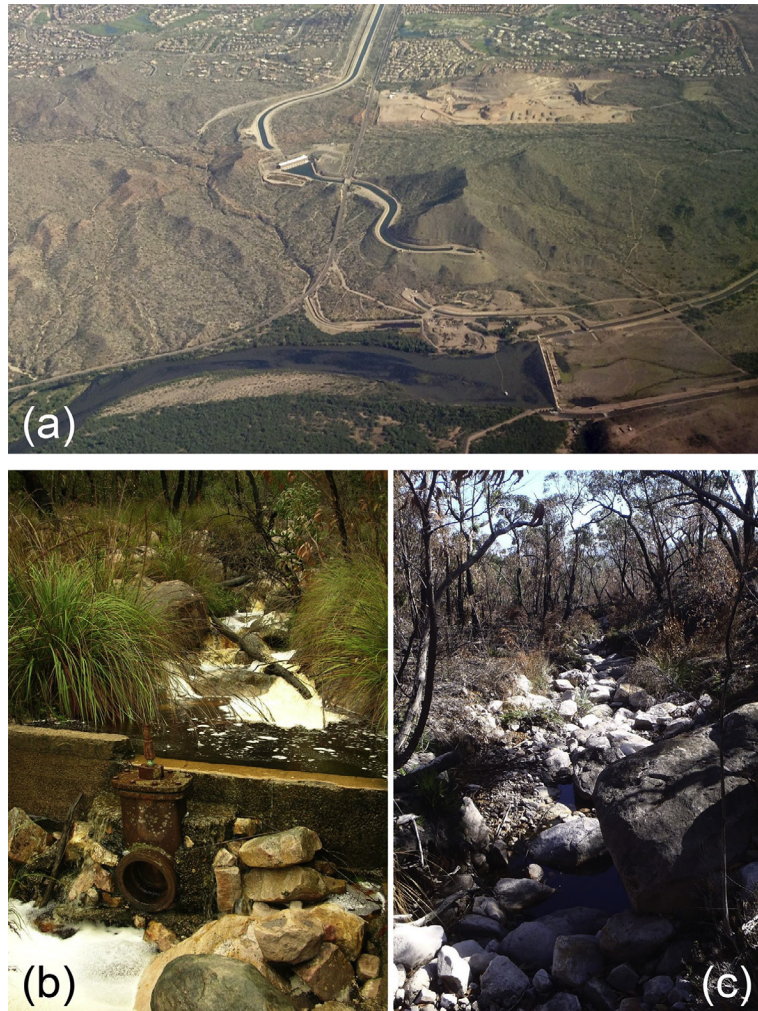
Genetic composition of species in IRES shows remarkable variation and has delivered some surprises in relation to their dispersal strategies. One unexpected pattern is that some species with very poor dispersal maintain either high genetic diversity, high effective population sizes, or both. For example, the Australian amphipod *Wangiannachiltonia guzikae* does not appear able to disperse even over short distances (<1 km), but has higher haplotype diversity and effective population size than other local species (Murphy et al., 2010). Similarly, the flightless giant water bug *Abedus herberti*, which occupies perennial pools of IRES in the southwest United States, maintains genetically distinct local populations, frequently with large effective sizes, that even show behavioral differences (Phillipsen and Lytle, 2013). The factors that enable species with poor dispersal to sustain high genetic diversity and large effective population sizes are unclear, but may include both life history traits (e.g., high fecundity, generalist diet) and intrinsic genetic factors (e.g., high mutation rate) (Murphy et al., 2010). In contrast, other IRES species with poor dispersal are strongly resistant and form small, isolated populations scattered across the landscape. This pattern may be more common in IRES than in perennial river systems and is documented for a variety of fish and invertebrate species (Murphy et al., 2010; Chester et al., 2015; Murphy et al., 2015). These findings demonstrate the importance of protecting isolated perennial waters as both ecological refuges and evolutionary refugia that may act as genetic diversity hotspots for some IRES species (Davis et al., 2013).

4.8.7 HUMAN IMPACTS IMPAIR RESISTANCE AND RESILIENCE PROCESSES

Anthropogenic alterations to IRES can dramatically impair the resistance and resilience processes of resident taxa. For example, water abstraction can increase habitat fragmentation, requiring aquatic taxa to disperse further between remaining wetted reaches. Physical disturbances, like gravel mining and off-road vehicle use (Fig. 4.3.7), may increase habitat fragmentation for terrestrial species inhabiting dry riverbeds and damage the desiccation-resistant resting stages of aquatic invertebrates. In this section, we explore just a few of the hydrological and physical alterations to IRES and how they affect resistance and resilience processes of resident taxa.

HYDROLOGICAL ALTERATIONS

Flow regulation by dams, weirs, and diversions has variable impacts that can either increase or decrease the amount of surface flow in IRES (Fig. 4.8.10; Kingsford et al., 2006; Reich et al., 2010; Chapter 5.1). Reduced population connectivity due to dams and weirs has been widely cited as impairing resilience processes for algae, aquatic invertebrates, and fish (Kingsford et al., 2006; Robson et al., 2008; Minckley and Marsh, 2009). For example, dams on IRES of the Great Plains (USA) blocked instream dispersal of several minnow species and reduced the resilience of their populations, causing extirpations from reaches above dams (Winston et al., 1991). However, taxa with resistance traits (e.g., crayfish) or the ability to disperse aerially (e.g., many aquatic insects) may exhibit stable populations in IRES regulated by weirs and dams, even while native fish are extirpated from the same streams (Chester et al., 2014). Passing “environmental flows” through small weirs in IRES can improve

**FIG. 4.8.10**

Diversions that reduce the volume and extent of flow in IRES. Granite Reef Dam (a) diverts nearly all of the water in the Salt River (USA) into a canal (upper center) which provides the Phoenix metropolitan area (2012 population: 4,329,534 inhabitants) with irrigation and drinking water. Similarly, but at a much smaller scale, a weir on Number 1 Creek (b) in Victoria, Australia, diverts nearly all flow from the creek, leaving only (c) small remnant pools in the channel downstream.

Photos: Courtesy M. Bogan (a) and E. Chester (b, c).

the dispersal and population connectivity of impacted taxa such as macroinvertebrates (Mackie et al., 2013). Terrestrial taxa that are not resistant to the inundation of dry riverbeds, or those that can only disperse through stream channels during dry periods, may benefit from reduced flow durations caused by water abstraction. However, other research suggests that stream drying reduces the diversity of terrestrial invertebrate taxa along riverbeds (McCluney and Sabo, 2012).

In contrast, artificially enhanced flows in IRES may increase the connectivity of formerly isolated reaches, which could increase the dispersal potential and resilience of populations. This alteration may benefit native species that prefer flowing water and are prone to dispersal via drift, such as some mayflies and caddisflies (Reich et al., 2010). However, artificially perennial flow can also facilitate the dispersal of nonnative fish species through IRES networks (Kingsford et al., 2006). Further research is needed to fully understand how artificially perennial flow regimes affect resistance and resilience processes in IRES.

The amount and duration of surface flow in IRES can be dramatically reduced by groundwater abstraction (Falke et al., 2011; Perkin et al., 2015). This abstraction can impair resistance processes in IRES by reducing the occurrence of remnant pools available for organisms to survive dry periods. For example, if groundwater pumping in the Arikaree River catchment (USA) continues to increase at current rates, only 36% of the already small number of remnant perennial pools are predicted to remain by 2045 (Falke et al., 2011). As well as reducing habitat in which aquatic individuals can resist dry periods, the loss of perennial pools will also impair resilience processes. The few remaining pools in the Arikaree River by 2045 will be concentrated in the lower reaches of the catchment, increasing the distance aquatic species will have to disperse during periods of flow.

PHYSICAL ALTERATIONS

Given the lower perceived value of IRES by most people, the amount of anthropogenic modification of these systems may be much higher than in perennial systems (Chapters 5.1 and 5.2). Dry streambeds are used as transportation corridors (e.g., seasonal roads and highways) in many regions, while off-road vehicles use is high in other regions. However, the physical alterations most likely to impair resistance and resilience processes in IRES are riparian vegetation clearing and riverbed gravel mining.

Riparian vegetation is commonly cleared along IRES in agricultural and urban areas. This clearing significantly reduces allochthonous organic matter biomass in IRES channels, depriving detritivores of essential food resources (Reid et al., 2008a,b). Such a loss of plant biomass is likely to reduce population resilience from stream drying, since these food resources are less available to colonists arriving from perennial refuges. However, this may not always be the case. For example, in southern Australia, large beds of submerged macrophytes often develop in agricultural IRES after flow resumes (Paice et al., 2016a). Carbon from these macrophyte stands can contribute substantially to aquatic food webs and may partially compensate for the loss of allochthonous resources (Paice et al., 2016b).

Riparian vegetation clearing may also reduce the resistance of aquatic taxa in dry riverbeds. Increased streambed temperature due to reduced canopy cover may affect the survival of desiccation-resistant invertebrates in dry streambeds. For example, temperatures in New Zealand streambeds climbed from ~18°C in forested reaches to ~40°C in cleared pasture reaches, and the abundance of aquatic invertebrates emerging from dormancy decreased by as much as 90% under higher temperature regimes (Storey and Quinn, 2013). However, in warmer biomes, even streambeds with intact riparian canopy can reach temperatures exceeding 60°C when dry (Steward et al., 2011), so dormant invertebrates in those IRES must tolerate higher temperatures.

Wholesale removal of dry riverbed sediments via gravel mining is a widespread disturbance to aquatic and riparian taxa, especially in dryland IRES like those of the southwestern United States (Bull and Scott, 1974). Gravel mining not only impacts the ability of fish and invertebrates to use and recolonize habitats (Brown et al., 1998) but can also remove or destroy desiccation-resistant resting

stages in dry sediment. The mining process itself, with large machinery driven through the streambed and damaging riparian vegetation, clearly would cause direct mortality of terrestrial taxa inhabiting dry streambeds. Furthermore, when extensive reaches are mined for gravel, in-channel habitat connectivity would be reduced, thus decreasing the resilience of both riparian and aquatic taxa. Our understanding of human impacts on dry streambeds, including those not considered here, such as biological pollution and the introduction of nonnative species, is still in its infancy. Each of these topics deserves further research attention.

4.8.8 KNOWLEDGE GAPS AND FUTURE RESEARCH DIRECTIONS

Although significant advances have been made in understanding resistance, resilience, and recovery processes in IRES, many unanswered questions remain. We know relatively little about the life history of most IRES species or the flexibility of different forms of dormancy in response to repeated wetting and drying cycles (Robson et al., 2011; Strachan et al., 2015; Stubbington et al., 2016). There have been few multiyear studies in IRES, limiting inferences about population structure and synchronization of life stages and recovery processes, among other themes. These are important because if we are to predict and manage the impacts of climate change and other disturbances in IRES, we must understand life histories, population dynamics, and the factors influencing them. Additionally, there is disparity in research focus among taxonomic groups. For example, aquatic invertebrates are relatively well studied, but primary producers and microbial taxa have received far less attention despite their supporting role in food webs and influence on recovery trajectories.

While the variety of refuges used by aquatic species to resist flow cessation in IRES is now quite well known, their spatial distribution and the interannual variability in their frequency, extent, and role in community recovery have not been well described. There is, for example, ongoing debate about the importance of the invertebrate seedbank in recovery of IRES assemblages, largely because rivers in different biomes appear to differ greatly in the abundance and richness of species emerging from river sediments (Chester and Robson, 2011; Stubbington and Datry, 2013). In addition to desiccation-resistant individuals, perennial refuges (e.g., remnant pools) are commonly cited as important sources of colonists to intermittent reaches when flow returns. But exactly how the spatial configuration of refuges, and community structure within those refuges, influences recolonization sequences in IRES following flow resumption has rarely been studied.

Resilience pathways can be difficult to study because it is hard to trace the movement of individuals, especially small taxa such as invertebrates and diatoms. Population genetics has helped greatly with understanding effective dispersal that leads to reproduction, but it cannot trace other movements that animals make (e.g., adult insects seeking refuge in caves or hollow trees when streams are dry). Many more measurements of actual movements by organisms are needed to fully understand resilience and recovery. Furthermore, new models for population structures unique to IRES may augment the models of population structure developed through studies of genetic patterns in perennial streams. Additional studies of multiple species with different lineages and life history strategies are needed to explore population structures across IRES.

Finally, alterations to flow regimes caused by climate change (Chapter 2.2) and prolonged drying exacerbated by water withdrawals will affect both terrestrial and aquatic dynamics, but more research is required to understand how these changes will alter IRES ecosystems. These data will be especially

important in guiding conservation and management decisions. For example, what are the best approaches for conserving resistant versus resilient species in IRES? When should managers focus on protecting connectivity among habitat patches versus preserving specific habitat patches that harbor endemic species (i.e., ecological refuges and evolutionary refugia, [Davis et al., 2013](#))? Is assisted dispersal a viable strategy for conserving natural communities in IRES with artificially extended dry periods or complete drying ([Lawler and Olden, 2011](#))? These will be pressing questions to address as flow regimes change in IRES because of increased human water demands and altered precipitation patterns in a globally changing climate.

REFERENCES

- Abell, D.L., 1984. Benthic invertebrates of some California intermittent streams. In: Jain, S., Moyle, P.B. (Eds.), *Vernal Pools and Intermittent Streams*. University of California, Davis, CA, pp. 46–60.
- Acuña, V., Munoz, I., Giorgi, A., Omella, M., Sabater, F., Sabater, S., 2005. Drought and postdrought recovery cycles in an intermittent Mediterranean stream: structural and functional aspects. *J. N. Am. Benthol. Soc.* 24, 919–933.
- Baguette, M., Blanchet, S., Legrand, D., Stevens, V.M., Turlure, C., 2013. Individual dispersal, landscape connectivity and ecological networks. *Biol. Rev.* 88, 310–326.
- Boersma, K.S., Lytle, D.A., 2014. Overland dispersal and drought-escape behavior in a flightless aquatic insect, *Abedus herberti* (Hemiptera: Belostomatidae). *Southwest. Nat.* 59, 301–302.
- Bogan, M.T., Boersma, K.S., 2012. Aerial dispersal of aquatic invertebrates along and away from arid-land streams. *Freshwater Sci.* 31, 1131–1144.
- Bogan, M.T., Lytle, D.A., 2007. Seasonal flow variation allows 'time-sharing' by disparate aquatic insect communities in montane desert streams. *Freshw. Biol.* 52, 290–304.
- Bogan, M.T., Lytle, D.A., 2011. Severe drought drives novel community trajectories in desert stream pools. *Freshw. Biol.* 56, 2070–2081.
- Bogan, M.T., Boersma, K.S., Lytle, D.A., 2013. Flow intermittency alters longitudinal patterns of invertebrate diversity and assemblage composition in an arid-land stream network. *Freshw. Biol.* 58, 1016–1028.
- Bogan, M.T., Boersma, K.S., Lytle, D.A., 2015. Resistance and resilience of invertebrate communities to seasonal and suprasedasonal drought in arid-land headwater streams. *Freshw. Biol.* 60, 2547–2558.
- Bonada, N., Rieradevall, M., Prat, N., Resh, V.H., 2006. Benthic macroinvertebrate assemblages and macrohabitat connectivity in Mediterranean-climate streams of northern California. *J. N. Am. Benthol. Soc.* 25, 32–43.
- Bostock, B.M., Adams, M., Laurenson, L.J.B., Austin, C.M., 2006. The molecular systematics of *Leiopotherapon unicolor* (Gunther, 1859): testing for cryptic speciation in Australia's most widespread freshwater fish. *Biol. J. Linn. Soc.* 87, 537–552.
- Boulton, A.J., Lake, P.S., 1992. The ecology of two intermittent streams in Victoria, Australia. III. Temporal changes in faunal composition. *Freshw. Biol.* 27, 123–138.
- Boulton, A.J., Lake, P.S., 2008. Effects of drought on stream insects and its ecological consequences. In: Lancaster, J., Briers, R.A. (Eds.), *Aquatic Insects: Challenges to Populations*. CABI International, Wallingford, pp. 91–102.
- Boulton, A.J., Peterson, C.G., Grimm, N.B., Fisher, S.G., 1992. Stability of an aquatic macroinvertebrate community in a multiyear hydrologic disturbance regime. *Ecology* 73, 2192–2207.
- Boumezzough, A., Musso, J.-J., 1983. Etude des communautés animales ripicoles du bassin de la rivière Aille (Var-France). I. Aspects biologiques et éco-éthologiques. *Ecologia Mediterranea* 9, 31–56.
- Briers, R.A., Gee, J.H.R., Cariss, H.M., Geoghegan, R., 2004. Inter-population dispersal by adult stoneflies detected by stable isotope enrichment. *Freshw. Biol.* 49, 425–431.

- Brown, A.V., Lyttle, M.M., Brown, K.B., 1998. Impacts of gravel mining on gravel bed streams. *Trans. Am. Fish. Soc.* 127, 979–994.
- Bull, W.B., Scott, K.M., 1974. Impact of mining gravel from urban stream beds in the southwestern United States. *Geology* 2, 171–174.
- Bunn, S.E., Hughes, J.M., 1997. Dispersal and recruitment in streams: evidence from genetic studies. *J. N. Am. Benthol. Soc.* 16, 338–346.
- Bunn, S.E., Thoms, M.C., Hamilton, S.K., Capon, S.J., 2006. Flow variability in dryland rivers: boom, bust and the bits in between. *River Res. Appl.* 22, 179–186.
- Cañedo-Argüelles, M., Boersma, K.S., Bogan, M.T., Olden, J.D., Phillipsen, I.C., Schriever, T.A., et al., 2015. Dispersal strength determines meta-community structure in a dendritic riverine network. *J. Biogeogr.* 42, 778–790.
- Chester, E.T., Robson, B.J., 2011. Drought refuges, spatial scale and recolonisation by invertebrates in non-perennial streams. *Freshw. Biol.* 56, 2094–2104.
- Chester, E.T., Robson, B.J., 2014. Do recolonization processes in intermittent streams have sustained effects on benthic algal density and assemblage composition? *Mar. Freshw. Res.* 65, 784–790.
- Chester, E.T., Matthews, T.G., Howson, T.J., Johnston, K., Mackie, J.K., Strachan, S.R., et al., 2014. Constraints upon the response of fish and crayfish to environmental flow releases in a regulated headwater stream network. *PLoS One* 9, e91925.
- Chester, E.T., Miller, A., Valenzuela, I., Wickson, S., Robson, B.J., 2015. Drought survival strategies, dispersal potential and persistence of invertebrate species in an intermittent stream landscape. *Freshw. Biol.* 60, 2066–2083.
- Clayton, D.A., 1993. Mudskippers. *Oceanogr. Mar. Biol. Annu. Rev.* 31, 507–577.
- Coetzee, C.G., 1969. The distribution of mammals in the Namib Desert and adjoining inland escarpment. *Sci. Pap. Namib Desert Res. Station* 40, 23–36.
- Collett, T., 1996. Insect navigation en route to the goal: multiple strategies for the use of landmarks. *J. Exp. Biol.* 199, 227–235.
- Colvin, R., Giannico, G.R., Li, J., Boyer, K.L., Gerth, W.J., 2009. Fish use of intermittent watercourses draining agricultural lands in the Upper Willamette River Valley, Oregon. *Trans. Am. Fish. Soc.* 138, 1302–1313.
- Copp, G.H., Faulkner, H., Doherty, S., Watkins, M.S., Majecki, J., 2002. Diel drift behaviour of fish eggs and larvae, in particular barbel, *Barbus barbus* (L.), in an English chalk stream. *Fish. Manag. Ecol.* 9, 95–103.
- Corti, R., Datry, T., 2012. Invertebrates and sestonic matter in an advancing wetted front travelling down a dry river bed (Albarine, France). *Freshwater Sci.* 31, 1187–1201.
- Corti, R., Datry, T., 2016. Terrestrial and aquatic invertebrates in the riverbed of an intermittent river: parallels and contrasts in community organisation. *Freshw. Biol.* 61, 1308–1320.
- Corti, R., Larned, S.T., Datry, T., 2013. A comparison of pitfall-trap and quadrat methods for sampling ground-dwelling invertebrates in dry riverbeds. *Hydrobiologia* 717, 13–26.
- Cover, M.R., Seo, J.H., Resh, V.H., 2015. Life history, burrowing behavior, and distribution of *Neohermes filicornis* (Megaloptera: Corydalidae), a long-lived aquatic insect in intermittent streams. *West. North Am. Nat.* 75, 474–490.
- Datry, T., 2012. Benthic and hyporheic invertebrate assemblages along a flow intermittence gradient: effect of duration of dry events. *Freshw. Biol.* 57, 565–574.
- Datry, T., 2017. Ecological effects of flow intermittence in gravel bed rivers. In: Tsutsumi, D., Laronne, J. (Eds.), *Gravel-Bed Rivers: Processes and Disasters*. Wiley, Chichester, pp. 261–298.
- Datry, T., Arscott, D.B., Sabater, S., 2011. Recent perspectives on temporary river ecology. *Aquat. Sci.* 73, 453–457.
- Datry, T., Corti, R., Philippe, M., 2012. Spatial and temporal aquatic-terrestrial transitions in the temporary Albarine River, France: responses of invertebrates to experimental rewetting. *Freshw. Biol.* 57, 716–727.
- Datry, T., Larned, S.T., Fritz, K.M., Bogan, M.T., Wood, P.J., Meyer, E.I., et al., 2014. Broad-scale patterns of invertebrate richness and community composition in temporary rivers: effects of flow intermittence. *Ecography* 37, 94–104.

- Datry, T., Bonada, N., Heino, J., 2016a. Towards understanding the organisation of metacommunities in highly dynamic ecological systems. *Oikos* 125, 149–159.
- Datry, T., Moya, N., Zubieta, J., Oberdorff, T., 2016b. Determinants of local and regional communities in intermittent and perennial headwaters of the Bolivian Amazon. *Freshw. Biol.* 61, 1335–1349.
- Davey, A.J.H., Kelly, D.J., 2007. Fish community responses to drying disturbances in an intermittent stream: a landscape perspective. *Freshw. Biol.* 52, 1719–1733.
- Davis, J., Pavlova, A., Thompson, R., Sunnucks, P., 2013. Evolutionary refugia and ecological refuges: key concepts for conserving Australian arid zone freshwater biodiversity under climate change. *Glob. Chang. Biol.* 19, 1970–1984.
- Del Rosario, R.B., Resh, V.H., 2000. Invertebrates in intermittent and perennial streams: is the hyporheic zone a refuge from drying? *J. N. Am. Benthol. Soc.* 19, 680–696.
- Descoux, S., Datry, T., Marmonier, P., 2013. Benthic and hyporheic invertebrate assemblages along a gradient of increasing streambed colmation by fine sediment. *Aquat. Sci.* 75, 493–507.
- Detenbeck, N.E., Devore, P.W., Niemi, G.J., Lima, A., 1992. Recovery of temperate-stream fish communities from disturbance: a review of case studies and synthesis of theory. *Environ. Manag.* 16, 33–53.
- Distefano, R.J., Magoulick, D.D., Imhoff, E.M., Larson, E.R., 2009. Imperiled crayfishes use hyporheic zone during seasonal drying of an intermittent stream. *J. N. Am. Benthol. Soc.* 28, 142–152.
- Driver, L.J., Hoeinghaus, D.J., 2016. Spatiotemporal dynamics of intermittent stream fish metacommunities in response to prolonged drought and reconnectivity. *Mar. Freshw. Res.* 67, 1667–1679.
- Drummond, L.R., McIntosh, A.R., Larned, S.T., 2015. Invertebrate community dynamics and insect emergence in response to pool drying in a temporary river. *Freshw. Biol.* 60, 1596–1612.
- Falke, J.A., Fausch, K.D., Magelky, R., Aldred, A., Durnford, D., Riley, L.K., et al., 2011. The role of groundwater pumping and drought in shaping ecological futures for stream fishes in a dryland river basin of the western Great Plains. *Ecohydrology* 4, 682–697.
- Flinders, C.A., Magoulick, D.D., 2003. Effects of stream permanence on crayfish community structure. *Am. Midl. Nat.* 149, 134–147.
- Fritz, K.M., Dodds, W.K., 2004. Resistance and resilience of macroinvertebrate assemblages to drying and flood in a tallgrass prairie stream. *Hydrobiologia* 527, 99–112.
- Gayraud, S., Philippe, M., 2001. Does subsurface interstitial space influence general features and morphological traits of the benthic macroinvertebrate community in streams? *Arch. Hydrobiol.* 151, 667–686.
- Glover, C.J.M., 1982. Adaptations of fishes in arid Australia. In: Barker, W.R., Greenslade, P.J.M. (Eds.), *Evolution of the Flora and Fauna of Arid Australia*. Peacock Publications, Adelaide, pp. 241–246.
- Gray, L.J., Fisher, S.G., 1981. Postflood recolonization pathways of macroinvertebrates in a lowland Sonoran Desert stream. *Am. Midl. Nat.* 106, 249–257.
- Greenwood, P.H., 1987. The natural history of African lungfishes. In: Bemis, W.E., Burggren, W.W., Kemp, N.E. (Eds.), *The Biology and Evolution of Lungfishes*. Alan R. Liss, New York, pp. 163–179.
- Greenwood, M.J., McIntosh, A.R., 2010. Low river flow alters the biomass and population structure of a riparian predatory invertebrate. *Freshw. Biol.* 55, 2062–2076.
- Hershkovitz, Y., Gasith, A., 2013. Resistance, resilience, and community dynamics in Mediterranean-climate streams. *Hydrobiologia* 719, 59–75.
- Hughes, J.M., Real, K.M., Marshall, J.C., Schmidt, D.J., 2012. Extreme genetic structure in a small-bodied freshwater fish, the purple-spotted gudgeon, *Mogurnda adspersa* (Eleotridae). *PLoS One* 7, e40546.
- Hwan, J.L., Carlson, S.M., 2016. Fragmentation of an intermittent stream during seasonal drought: intra-annual and interannual patterns and biological consequences. *River Res. Appl.* 32, 856–870.
- Jaeger, K.L., Olden, J.D., 2012. Electrical resistance sensor arrays as a means to quantify longitudinal connectivity of rivers. *River Res. Appl.* 28, 1843–1852.
- Johnston, K., Robson, B.J., Austin, C.M., 2008. Population structure and life history characteristics of *Euastacus bispinosus* and *Cherax destructor* (Parastacidae) in the Grampians National Park, Australia. *Freshwater Crayfish* 16, 165–173.

- Kawanishi, R., Inoue, M., Dohi, R., Fujii, A., Miyake, Y., 2013. The role of the hyporheic zone for a benthic fish in an intermittent river: a refuge, not a graveyard. *Aquat. Sci.* 75, 425–431.
- Kerecsy, A., Balcombe, S.R., Tischler, M., Arthington, A.H., 2013. Fish movement strategies in an ephemeral river in the Simpson Desert, Australia. *Austral Ecol.* 38, 798–808.
- Kingsford, R.T., Lemly, A.D., Thompson, J.R., 2006. Impacts of dams, river management and diversions on desert rivers. In: Kingsford, R.T. (Ed.), *Ecology of Desert Rivers*. Cambridge University Press, Cambridge, pp. 203–247.
- Kingsley, K.J., 1985. *Eretes sticticus* (L.) (Coleoptera: Dytiscidae): life history observations and an account of a remarkable event of synchronous emigration from a temporary desert pond. *Coleopt. Bull.* 39, 7–10.
- Kriska, G., Bernáth, B., Farkas, R., Horváth, G., 2009. Degrees of polarization of reflected light eliciting polarotaxis in dragonflies (Odonata), mayflies (Ephemeroptera) and tabanid flies (Tabanidae). *J. Insect Physiol.* 55, 1167–1173.
- Labbe, T.R., Fausch, K.D., 2000. Dynamics of intermittent stream habitat regulate persistence of a threatened fish at multiple scales. *Ecol. Appl.* 10, 1774–1791.
- Lake, P.S., 2000. Disturbance, patchiness, and diversity in streams. *J. N. Am. Benthol. Soc.* 19, 575–592.
- Lake, P.S., 2011. *Drought and Aquatic Ecosystems: Effects and Responses*. Wiley Blackwell, Chichester.
- Lambeets, K., Vandegehuchte, M.L., Maelfait, J.P., Bonte, D., 2008. Understanding the impact of flooding on trait-displacements and shifts in assemblage structure of predatory arthropods on river banks. *J. Anim. Ecol.* 77, 1162–1174.
- Larimore, R.W., Childers, W.F., Heckrotte, C., 1959. Destruction and re-establishment of stream fish and invertebrates affected by drought. *Trans. Am. Fish. Soc.* 88, 261–285.
- Larned, S.T., Datry, T., Robinson, C.T., 2007. Invertebrate and microbial responses to inundation in an ephemeral river reach in New Zealand: effects of preceding dry periods. *Aquat. Sci.* 69, 554–567.
- Lawler, J.J., Olden, J.D., 2011. Reframing the debate over assisted colonization. *Front. Ecol. Environ.* 9, 569–574.
- Leigh, C., Bonada, N., Boulton, A.J., Hugueny, B., Larned, S.T., Vander Vorste, R., et al., 2016. Invertebrate assemblage responses and the dual roles of resistance and resilience to drying in intermittent streams. *Aquat. Sci.* 78, 291–301.
- Lytle, D.A., 1999. Use of rainfall cues by *Abedus herberti* (Hemiptera: Belostomatidae): a mechanism for avoiding flash floods. *J. Insect Behav.* 12, 1–12.
- Lytle, D.A., McMullen, L.E., Olden, J.D., 2008. Drought-escape behaviors of aquatic insects may be adaptations to highly variable flow regimes characteristic of desert rivers. *Southwest. Nat.* 53, 399–402.
- Mackie, J.K., Chester, E.T., Matthews, T.G., Robson, B.J., 2013. Macroinvertebrate response to environmental flows in headwater streams in western Victoria, Australia. *Ecol. Eng.* 53, 100–105.
- Marshall, J.C., Menke, N., Crook, D.A., Lobegeiger, J.S., Balcombe, S.R., Huey, J.A., et al., 2016. Go with the flow: the movement behaviour of fish from isolated waterhole refugia during connecting flow events in an intermittent dryland river. *Freshw. Biol.* 61, 1242–1258.
- McArthur, J.V., Barnes, J.R., 1985. Patterns of macroinvertebrate colonization in an intermittent Rocky Mountain stream in Utah. *Great Basin Nat.* 45, 117–123.
- McCluney, K.E., Sabo, J.L., 2012. River drying lowers the diversity and alters the composition of an assemblage of desert riparian arthropods. *Freshw. Biol.* 57, 91–103.
- McMaster, D., Bond, N., 2008. A field and experimental study on the tolerances of fish to *Eucalyptus camaldulensis* leachate and low dissolved oxygen concentrations. *Mar. Freshw. Res.* 59, 177–185.
- Minckley, W.L., Barber, W.E., 1971. Some aspects of the biology of the longfin dace, a cyprinid fish characteristic of streams of the Sonoran Desert. *Southwest. Nat.* 15, 459–464.
- Minckley, W.L., Marsh, P.C., 2009. *Inland Fishes of the Greater Southwest: Chronicle of a Vanishing Biota*. The University of Arizona Press, Tucson.
- Morán-Ordóñez, A., Pavlova, A., Pinder, A.M., Sim, L., Sunnucks, P., Thompson, R.M., et al., 2015. Aquatic communities in arid landscapes: local conditions, dispersal traits and landscape configuration determine local biodiversity. *Divers. Distrib.* 21, 1230–1241.

- Mossop, K.D., Adams, M., Unmack, P.J., Smith Date, K.L., Wong, B.B.M., Chapple, D.G., 2015. Dispersal in the desert: ephemeral water drives connectivity and phylogeography of an arid-adapted fish. *J. Biogeogr.* 42, 2374–2388.
- Murphy, N.P., Guzik, M.T., Worthington-Wilmer, J., 2010. The influence of landscape on population structure of four invertebrates in groundwater springs. *Freshw. Biol.* 55, 2499–2509.
- Murphy, A.L., Pavlova, A., Thompson, R., Davis, J., Sunnucks, P., 2015. Swimming through sand: connectivity of aquatic fauna in deserts. *Ecol. Evol.* 5, 5252–5264.
- Nimmo, D.G., Mac Nally, R., Cunningham, S.C., Haslem, A., Bennett, A.F., 2015. Vive la résistance: reviving resistance for 21st century conservation. *Trends Ecol. Evol.* 30, 516–523.
- Paetzold, A., Bernet, J.F., Tockner, K., 2006. Consumer-specific responses to riverine subsidy pulses in a riparian arthropod assemblage. *Freshw. Biol.* 51, 1103–1115.
- Paice, R.L., Chambers, J.M., Robson, B.J., 2016a. Native submerged macrophyte distribution in seasonally-flowing, south-western Australian streams in relation to stream condition. *Aquat. Sci.* <http://dx.doi.org/10.1007/s00027-016-0488-x>.
- Paice, R.L., Chambers, J.M., Robson, B.J., 2016b. Potential of submerged macrophytes to support food webs in lowland agricultural streams. *Mar. Freshw. Res.* <http://dx.doi.org/10.1071/MF15391>.
- Paltridge, R.M., Dostine, P.L., Humphrey, C.L., Boulton, A.J., 1997. Macroinvertebrate recolonization after rewetting of a tropical seasonally-flowing stream (Magela Creek, Northern Territory, Australia). *Mar. Freshw. Res.* 48, 633–645.
- Perkin, J.S., Gido, K.B., Costigan, K.H., Daniels, M.D., Johnson, E.R., 2015. Fragmentation and drying ratchet down Great Plains stream fish diversity. *Aquat. Conserv. Mar. Freshwat. Ecosyst.* 25, 639–655.
- Phillipsen, I.C., Lytle, D.A., 2013. Aquatic insects in a sea of desert: population genetic structure is shaped by limited dispersal in a naturally fragmented landscape. *Ecography* 36, 731–743.
- Phillipsen, I.C., Kirk, E.H., Bogan, M.T., Mims, M.C., Olden, J.D., Lytle, D.A., 2015. Dispersal ability and habitat requirements determine landscape-level genetic patterns in desert aquatic insects. *Mol. Ecol.* 24, 54–69.
- Pires, D.F., Beja, P., Magalhaes, M.F., 2014. Out of pools: movement patterns of Mediterranean stream fish in relation to dry season refugia. *River Res. Appl.* 30, 1269–1280.
- Poff, N.L., Ward, J.V., 1989. Implications of streamflow variability and predictability for lotic community structure: a regional analysis of streamflow patterns. *Can. J. Fish. Aquat. Sci.* 46, 1805–1818.
- Razeng, E., Morán-Ordóñez, A., Brim Box, J., Thompson, R., Davis, J., Sunnucks, P., 2016. A potential role for overland dispersal in shaping aquatic invertebrate communities in arid regions. *Freshw. Biol.* 61, 745–757.
- Reich, P., McMaster, D., Bond, N., Metzeling, L., Lake, P.S., 2010. Examining the ecological consequences of restoring flow intermittency to artificially perennial lowland streams: patterns and predictions from the Broken-Boosey Creek system in Northern Victoria, Australia. *River Res. Appl.* 26, 529–545.
- Reid, D.J., Lake, P.S., Quinn, G.P., Reich, P., 2008a. Association of reduced riparian vegetation cover in agricultural landscapes with coarse detritus dynamics in lowland streams. *Mar. Freshw. Res.* 59, 998–1014.
- Reid, D.J., Quinn, G.P., Lake, P.S., Reich, P., 2008b. Terrestrial detritus supports the food webs in lowland intermittent streams of south-eastern Australia: a stable isotope study. *Freshw. Biol.* 53, 2036–2050.
- Robson, B.J., Matthews, T.G., Lind, P.R., Thomas, N.A., 2008. Pathways for algal recolonization in seasonally-flowing streams. *Freshw. Biol.* 53, 2385–2401.
- Robson, B.J., Chester, E.T., Austin, C.M., 2011. Why life history information matters: drought refuges and macroinvertebrate persistence in non-perennial streams subject to a drier climate. *Mar. Freshw. Res.* 62, 801–810.
- Robson, B.J., Chester, E.T., Mitchell, B.D., Matthews, T.G., 2013. Disturbance and the role of refuges in Mediterranean climate streams. *Hydrobiologia* 719, 77–91.
- Rosado, J., Morais, M., Tockner, K., 2015. Mass dispersal of terrestrial organisms during first flush events in a temporary stream. *River Res. Appl.* 31, 912–917.

- Sánchez-Montoya, M.D.M., Von Schiller, D., Ruhí, A., Pechar, G.S., Proia, L., Miñano, J., et al., 2015. Responses of ground-dwelling arthropods to surface flow drying in channels and adjacent habitats along Mediterranean streams. *Ecohydrology* 9, 1376–1387.
- Segev, O., Blaustein, L., 2014. Influence of water velocity and predation risk on fire salamander (*Salamandra atra*) larval drift among temporary pools in ephemeral streams. *Freshwater Sci.* 33, 950–957.
- Sheldon, F., Bunn, S.E., Hughes, J.M., Arthington, A.H., Balcombe, S.R., Fellows, C.S., 2010. Ecological roles and threats to aquatic refugia in arid landscapes: dryland river waterholes. *Mar. Freshw. Res.* 61, 885–895.
- Stanley, E.H., Buschman, D.L., Boulton, A.J., Grimm, N.B., Fisher, S.G., 1994. Invertebrate resistance and resilience to intermittency in a desert stream. *Am. Midl. Nat.* 131, 288–300.
- Stebbins, R.C., 2003. *A Field Guide to Western Reptiles and Amphibians*, third ed. Houghton Mifflin Harcourt, Boston, MA.
- Stevens, L.E., Polhemus, J.T., Durfree, R.S., Olson, C.A., 2007. Large mixed-species dispersal flights of predatory and scavenging aquatic Heteroptera and Coleoptera, northern Arizona, USA. *West. North Am. Nat.* 67, 587–592.
- Steward, A.L., Marshall, J.C., Sheldon, F., Harch, B., Choy, S., Bunn, S.E., et al., 2011. Terrestrial invertebrates of dry river beds are not simply subsets of riparian assemblages. *Aquat. Sci.* 73, 551–566.
- Steward, A.L., Von Schiller, D., Tockner, K., Marshall, J.C., Bunn, S.E., 2012. When the river runs dry: human and ecological values of dry riverbeds. *Front. Ecol. Environ.* 10, 202–209.
- Storey, R.G., Quinn, J.M., 2013. Survival of aquatic invertebrates in dry bed sediments of intermittent streams: temperature tolerances and implications for riparian management. *Freshwater Sci.* 32, 250–266.
- Strachan, S.R., Chester, E.T., Robson, B.J., 2015. Freshwater invertebrate life history strategies for surviving desiccation. *Springer Sci. Rev.* 3, 57–75.
- Stubbington, R., 2012. The hyporheic zone as an invertebrate refuge: a review of variability in space, time, taxa and behaviour. *Mar. Freshw. Res.* 63, 293–311.
- Stubbington, R., Datry, T., 2013. The macroinvertebrate seedbank promotes community persistence in temporary rivers across climate zones. *Freshw. Biol.* 58, 1202–1220.
- Stubbington, R., Gunn, J., Little, S., Worrall, T.P., Wood, P.J., 2016. Macroinvertebrate seedbank composition in relation to antecedent duration of drying and multiple wet-dry cycles in a temporary stream. *Freshw. Biol.* 61, 1293–1307.
- Taylor, C.M., Warren, M.L., 2001. Dynamics in species composition of stream fish assemblages: environmental variability and nested subsets. *Ecology* 82, 2320–2330.
- Thompson, G.G., Withers, P.C., 2002. Aerial and aquatic respiration of the Australian desert goby, *Chlamydogobius eremius*. *Comp. Biochem. Physiol. A Mol. Integr. Physiol.* 131, 871–879.
- Timoner, X., Acuña, V., Frampton, L., Pollard, P., Sabater, S., Bunn, S.E., 2014. Biofilm functional responses to the rehydration of a dry intermittent stream. *Hydrobiologia* 727, 185–195.
- Vadher, A.N., Stubbington, R., Wood, P.J., 2015. Fine sediment reduces vertical migrations of *Gammarus pulex* (Crustacea: Amphipoda) in response to surface water loss. *Hydrobiologia* 753, 61–71.
- Vander Vorste, R., Malard, F., Datry, T., 2016a. Is drift the primary process promoting the resilience of river invertebrate communities? A manipulative field experiment in an intermittent alluvial river. *Freshw. Biol.* 61, 1276–1292.
- Vander Vorste, R., Mermillod-Blondin, F., Hervant, F., Mons, R., Forcellini, M., Datry, T., 2016b. Increased depth to the water table during river drying decreases the resilience of *Gammarus pulex* and alters ecosystem function. *Ecohydrology* 9, 1177–1186.
- Velasco, J., Millan, A., 1998. Insect dispersal in a drying desert stream: effects of temperature and water loss. *Southwest. Nat.* 43, 80–87.
- Verberk, W.C.E.P., Sipel, H., Esselink, H., 2008. Life history strategies in freshwater macroinvertebrates. *Freshw. Biol.* 53, 1722–1738.

- Whitney, J.E., Gido, K.B., Martin, E.C., Hase, K.J., 2016. The first to arrive and the last to leave: colonisation and extinction dynamics of common and rare fishes in intermittent prairie streams. *Freshw. Biol.* 61, 1321–1334.
- Wickson, S.J., Chester, E.T., Robson, B.J., 2012. Aestivation provides flexible mechanisms for survival of stream drying in a larval trichopteran (Leptoceridae). *Mar. Freshw. Res.* 63, 821–826.
- Wigington Jr., P.J., Ebersole, J.L., Colvin, M.E., Leibowitz, S.G., Miller, B., Hansen, B., et al., 2006. Coho salmon dependence on intermittent streams. *Front. Ecol. Environ.* 4, 513–518.
- Williams, D.D., 1977. Movements of benthos during the recolonization of temporary streams. *Oikos* 29, 306–312.
- Williams, D.D., 2006. *The Biology of Temporary Waters*. Oxford University Press, Oxford.
- Williams, D.D., Hynes, H.B., 1976. The recolonization mechanisms of stream benthos. *Oikos* 27, 265–272.
- Winston, M.R., Taylor, C.M., Pigg, J., 1991. Upstream extirpation of four minnow species due to damming of a prairie stream. *Trans. Am. Fish. Soc.* 120, 98–105.

FURTHER READING

- Stubbington, R., Boulton, A.J., Little, S., Wood, P.J., 2015. Changes in invertebrate assemblage composition in benthic and hyporheic zones during a severe suprasedasonal drought. *Freshwater Sci.* 34, 344–354.